Recommendation and Information Diffusion on Epinions

SNTP Assignment 2 — Social Network Theory and Programming

Ahmad Fareed | 23F3008 | NU-FAST Chiniot-Faisalabad | 2026-10-21

Measured artifacts: results/fast_demo/ produced by configs/fast_demo.yaml. The full-scale configuration configs/default.yaml was not executed; no full-scale number is claimed anywhere in this report. Every diffusion, intervention and satisfaction outcome is a simulation, not an observed adoption.

Abstract

This report studies recommendation and information diffusion on the Epinions who-trusts-whom network: 49,289 users, 139,738 items, 664,824 ratings and 487,002 trust edges (sparsity 0.999903). A training/validation/test split (70/15/15, seed 42) drives item-based collaborative filtering, matrix factorisation and a trust-aware social extension. Diffusion is studied with canonical and preference-aware Independent Cascade and with SIR on a 9,883-user, 301,710-edge 2-hop ego subgraph. Seed selection compares random, degree and greedy/CELF strategies; group recommendation compares average, least-misery and most-pleasure aggregation; a relevance intervention trades simulated spread against a predicted-satisfaction proxy. All Monte-Carlo comparisons use genuine common random outcomes (one per-edge threshold per trial, reused across seed sets, probabilities and variants), which makes paired differences and the CELF lazy bound meaningful. Headline results: MF reaches test MAE 0.8025 and the social model test nDCG@10 0.00518; canonical ICM at p=0.05 averages 4,695.5 adopters versus 694.3 for the preference-aware variant; greedy influence maximisation gains 5.6 adopters over degree centrality inside a restricted 25-user pool; and CELF matches greedy exactly with 24 versus 33 evaluations. Negative and weak results are reported as measured.

1. Data, Split and Protocol

The Epinions transfer contains 664,824 ratings (49,289 users, 139,738 items) and 487,002 directed trust edges after removing 181 self-loops. 40,162 users both rate and trust; 105,966 trust edges are reciprocal. Ratings are skewed positive (301,053 fives versus 43,228 ones).

Split	Ratings	Share	Notes
Train	463,663	70%	Items/users for fitting
Validation	100,603	15%	Model/lambda/target selection
Test	100,558	15%	Final observed evaluation only

Seed 42 is fixed. Users with fewer than three ratings are excluded from evaluation (11,676 users); 28,487 remain. The target item for diffusion and intervention is chosen from a training-only popularity band, and the choice is recorded with the band size and the target's training rating count. Unobserved items are never treated as negative examples.

2. Recommendation Models (R1–R3)

2.1 Item-based CF, matrix factorisation and the social extension

Item-based CF uses cosine item similarity over the training matrix (max_neighbors=20, min_corated=2). Matrix factorisation uses 20 factors, lr=0.01, reg=0.005, 20 epochs. The social model blends the MF score with trust-neighbour and similar-user terms; lambda was selected on validation only (chosen λ = 0.3).

Model	MAE	RMSE	nDCG@10	Cov.	Uniq.	Div.
Item-based CF	1.0740	1.4519	0.00163	0.01715	2,396	0.874
Matrix factorisation	0.8025	1.0510	0.00314	0.00197	275	0.970
Social (trust, λ=0.3)	0.8008	1.0479	0.00518	0.01046	1,461	0.965

MAE/RMSE use the full model; nDCG/coverage use the raw ranking scores. The social model is best on every observed accuracy metric here. Item-CF has the highest coverage because its similarity neighbourhood spreads recommendations across more of the catalogue, but its ranking quality is the weakest.

2.2 MF ranking trade-off (disclosed)

The submitted MF top-N lists originally collapsed onto a few popular items because the global item bias dominated the within-user score variance (bias share 0.99913). Reducing the bias weight to 0.05, selected on validation, widens catalogue coverage. The trade-off is real and is reported rather than hidden:

- Validation nDCG@10 rose from 0.00266 (15 unique items) to 0.00711 (278 unique items).
- On test, the selected model reaches nDCG@10 0.00314 with 275 unique items, versus the previously committed configuration's test nDCG@10 0.00343 over ~16 unique items (coverage 0.0001145 × 139,738 ≈ 16). Coverage rose roughly 17× while test nDCG@10 fell by about 8.5%.
- Validation metrics, not test metrics, drove the choice; test was read once for the final table.

2.3 Trust versus similarity

On 165 held-out ratings from 50 sampled users (λ=0.3, k=10), trust neighbours give RMSE 1.0701 and similar unconnected users give 1.0652, versus baseline 1.0664. The two tie in practice, so this sample does not establish that trust beats similarity.

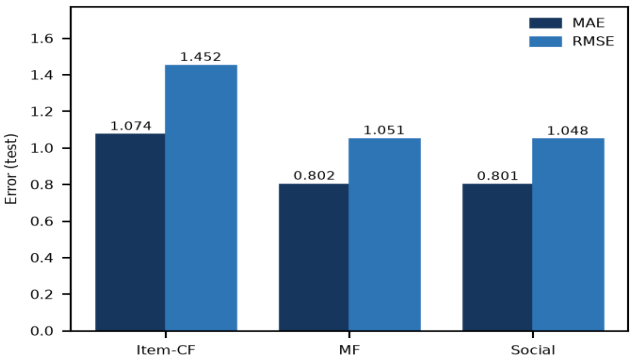


Fig. 1. Observed test prediction error per recommender (R7). Lower is better.

3. Information Diffusion (R4)

Diffusion runs on a documented 2-hop ego subgraph (centre user 0): 9,883 users and 301,710 directed propagation edges. Canonical ICM activates each out-edge once with probability p; preference-aware ICM separates exposure from adoption, where the adoption probability is scaled by the recommender's predicted affinity. SIR adds a recovery chance γ per round. Every setting uses 10 Monte-Carlo trials with the same seed sets.

3.1 Common random outcomes (CRN)

A sequential RNG restarted at the same integer seed still yields different live edges when the cascade path changes. This work generates one activation threshold per directed edge per trial and reuses it across every seed set, probability and ICM variant; preference-aware ICM adds one adoption threshold per user; SIR uses one transmission threshold per (edge, round) and one recovery threshold per (user, round). Consequently the finite-trial objective is deterministic, monotone and submodular, paired differences are meaningful, and the CELF lazy bound is valid. Thresholds are not shared across different graphs because edge identities differ (Thresholds are shared within a trial; separate trials are independent. Thresholds are not shared across different graphs because edge identities differ.).

Model	Setting	Mean	SD	Rounds
ICM canonical	p=0.01	69.9	111.6	6.9
ICM canonical	p=0.05	4,695.5	49.3	11.0
Preference-aware	p=0.01	9.3	6.1	2.6
Preference-aware	p=0.05	694.3	359.3	11.4
SIR	β=0.01, γ=0.0	2,271.7	982.8	25.0
SIR	β=0.01, γ=0.2	198.7	304.3	15.6
SIR	β=0.05, γ=0.0	8,277.7	97.9	25.0
SIR	β=0.05, γ=0.2	6,133.4	64.1	25.0

Increasing p raises ICM spread strongly. Preference-aware adoption cuts the cascade because low-affinity users often do not adopt, even when they are exposed. SIR with γ=0.05 never recovers users in the 25-round horizon (SI-like); γ=0.2 sharply reduces the infected population at low β, showing that recovery matters most when transmission is weak.

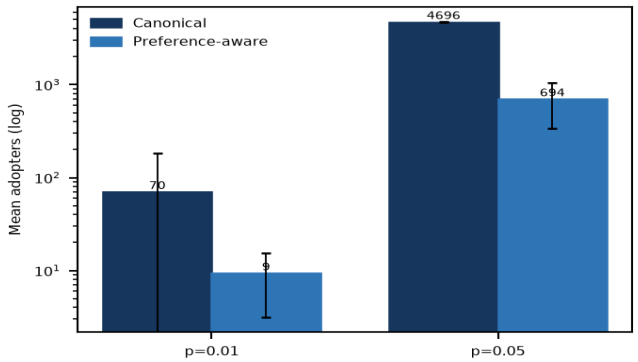


Fig. 2. Rare-target (item 3, one training rating) ICM spread on the 9,883-user subgraph, 10 trials, error bars = SD.

3.2 Target popularity and network structure

The same models are repeated on a moderate target (item 69, nine training ratings, band size 10,807) with its own seed sets. Canonical spread is similar across the two targets; preference-aware spread drops much more on the moderate target at p=0.05 (106.5 versus 694.3). The affinity function, not the target's raw popularity, drives the preference-aware differences.

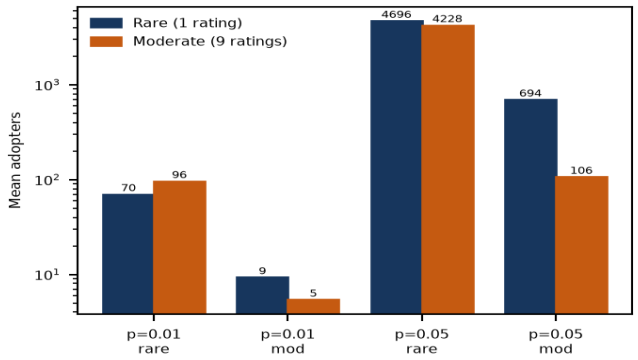


Fig. 3. Rare versus moderate target item, same graph and seeds (R4). Canonical pairs are the two left bars of each p; preference-aware the two right bars.

Setting	Observed	Randomized	Change	SD obs/rand
p=0.01	69.9	12.3	-82.4%	111.6 / 8.1
p=0.05	4,695.5	4,754.4	+1.3%	49.3 / 51.8

A degree-preserving directed randomization keeps node count, edge count and the in/out-degree sequence identical (521,458 accepted swaps). At p=0.01 the observed arrangement spreads far more (69.9 versus 12.3, -82.4%); at p=0.05 the two structures are statistically indistinguishable. Structure matters at the sparse-propagation regime, not at saturation.

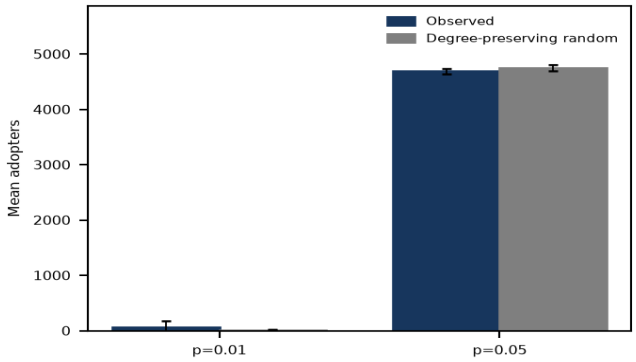


Fig. 4. Observed versus degree-preserving randomized structure, same seed sets, 10 trials (R4).

4. Influence Maximization (R5)

Random, degree-centrality and greedy seed selection are compared on the same graph at budgets [1, 3, 5], p=0.05, 5 Monte-Carlo trials per evaluation and 20 random trials. Because a full greedy sweep over 9,883 users is infeasible here, all three methods draw from the same restricted pool of 25 highest out-degree users (top out-degree users in the propagation graph, ties by ascending user index). The result therefore describes that pool, not the whole graph.

Budget	Random	Degree	Greedy	Greedy−deg	Greedy−rand
1	4,719.4	4,718.8	4,718.8	0.0	-0.6
3	4,719.4	4,718.8	4,724.4	5.6	5.0
5	4,719.6	4,718.8	4,724.4	5.6	4.8

The graph is near saturation: one seed already reaches about 4,719 users. Greedy is not worse than the baselines and gains up to 5.6 adopters over degree and 4.8 over random at budget 5, but the lift is about 0.12%. Random is marginally best at budget 1. The honest reading is that the restricted high-degree pool leaves little room for improvement, not that greedy is useless.

Marginal gains are non-negative by construction under CRN, which removes the previous artefact where a freshly drawn random stream produced a negative gain. Greedy required 115 evaluations and 45.1 s of selection time; evaluating all three methods on common trials cost 1.0 s (degree) and 0.9 s (greedy).

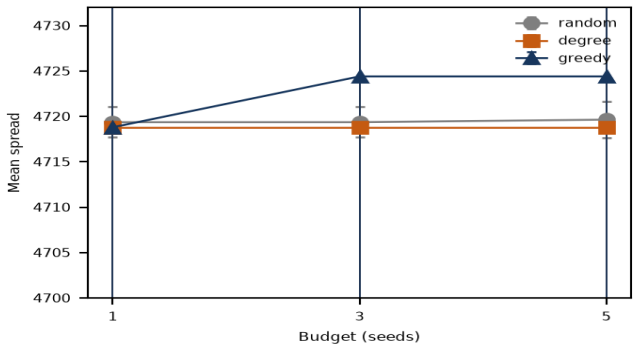


Fig. 5. Mean simulated spread versus budget for random, degree and greedy selection on the restricted 25-user pool (R5); error bars = SD across evaluation trials.

5. Group Recommendation (R6)

3 conflicting-preference groups are built from the training split; each group's top-N list is produced under average, least-misery and most-pleasure aggregation. Ordering uses the recommender's raw (unclipped) aggregate so items that clip to the same rating no longer tie; the reported group score is the clipped aggregate satisfaction. Spread for the top item is a preference-aware ICM simulation from the group members as seeds.

Strategy	Member sat.	Group score	Raw rank	Spread
average	4.915	4.915	5.180	713.0
least misery	4.915	4.717	4.717	713.0
most pleasure	4.915	5.000	5.636	713.0

Least-misery returns the lowest group score by definition; most-pleasure returns 5.0 because at least one member predicts the ceiling, yet its raw ranking score is distinct (5.904 for group 2), which is why the ordering is no longer an artefact of clipping. All three strategies select the same top item in every group in this run, so the simulated spread is identical; that is a property of these training-only groups, not evidence that the strategies are equivalent.

6. Comprehensive Evaluation (R7)

Observed test ranking is paired with a preference-aware ICM simulation that uses each model's affinity function on the same target item, seed sets and CRN thresholds. The two answer different questions and are not interchangeable.

Model	nDCG@10	RMSE	Adopt.	SD	Sat. proxy
item based cf	0.00163	1.4519	336.2	405.2	4.118
matrix factorization	0.00314	1.0510	547.2	445.8	4.304
social recommender	0.00518	1.0479	744.2	374.2	4.322

The best observed nDCG model and the largest simulated spread both belong to the social recommender (social recommender, spread 744.2), so the two rankings agree in this run. The simulated cascade uses one target item from a training popularity band and is not a platform-wide forecast. Coverage and diversity depend on the evaluated list sample (500 seeded users) and on training-derived item features.

7. Intervention, CELF and Cost Scaling (R8)

7.1 Relevance intervention

A minimum predicted-affinity threshold blocks low-affinity adoption inside the simulator (target item 3, 10 trials, same CRN thresholds). Spread falls as the threshold rises; the adopter-rating proxy is a simulator-side quantity, not observed satisfaction.

Threshold	Mean adopters	Change	Rating proxy	≥4 share
0.0	668.7	0.0%	4.308	0.838
0.3	668.7	0.0%	4.308	0.838
0.6	655.5	-2.0%	4.317	0.844
0.8	106.9	-84.0%	4.494	0.989
0.9	7.1	-98.9%	4.391	0.831

Thresholds at or below every observed affinity (0.0, 0.3) change nothing. At 0.6 spread barely moves; at 0.8 it falls 84% while the rating proxy rises; at 0.9 the cascade is almost fully suppressed (7.1 adopters) and the proxy falls because the remaining adopters are not the highest-affinity users. The measured table, not an assumed monotone trade-off, is the evidence.

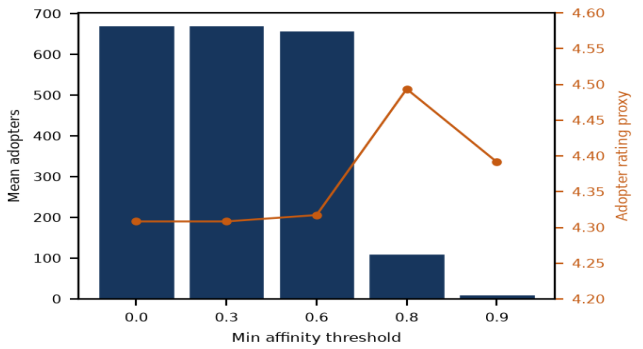


Fig. 6. Simulated spread (bars, left axis) and adopter-satisfaction proxy (line, right axis) versus minimum-affinity threshold (R8).

7.2 CELF lazy greedy

With the CRN objective now deterministic and submodular, CELF's lazy bound is valid and should reproduce standard greedy exactly. It does:

Variant	Seeds	Evaluations	Runtime (s)	Marginal gains
Standard greedy	[494, 321, 22]	33	61.30	4,750.4, 3.8, 0.0
CELF greedy	[494, 321, 22]	24	56.69	4,750.4, 3.8, 0.0

CELF used 27.27% fewer evaluations (24 vs 33) and ran 1.08× faster in this measured setting, selecting the identical seed set. A separate controlled check at p=1.0 on a 120-node graph also produced identical seeds and equal evaluation counts (39 each).

7.3 Cost scaling

Dimension	Size	Runtime	Evals	Memory peak
Network 25%	2,471 nodes	3.01 s	29	3.07 MB
Network 50%	4,942 nodes	9.90 s	29	6.92 MB
Items 25%	34,934 items	1.057 ms/user	—	1.07 MB
Items 50%	69,869 items	2.017 ms/user	—	2.13 MB
Items 100%	139,738 items	6.523 ms/user	—	4.26 MB

Greedy runtime grows super-linearly with network size because each candidate evaluation is a full Monte-Carlo cascade; item scoring is linear in catalogue size. The scaling runs use a single Monte-Carlo trial to isolate cost, so they are cost measurements, not spread estimates.

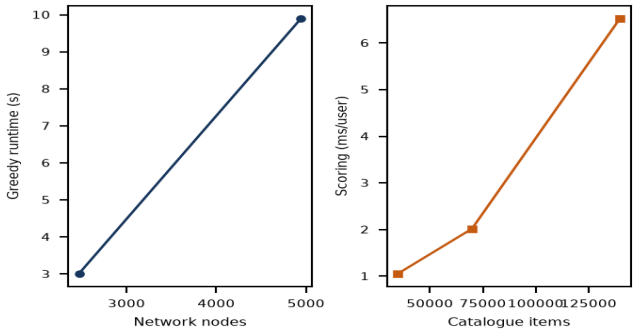


Fig. 7. Cost scaling: greedy network runtime (left) and MF scoring cost (right). One Monte-Carlo trial, cost isolated (R8).

8. Reproducibility, Limitations and Conclusion

8.1 Reproducibility

- Measured fast-demo pipeline: 473.6 s of stage time (7.9 min) on the development machine. Longest stages: MF training 120.2 s, R5 69.7 s, R8 140.0 s.
- Commands: python -m pytest -q (103 tests, ~1.7 s) and python -m sntp.cli run --config configs/fast_demo.yaml. The report is regenerated with python scripts/generate_report.py --results results/fast_demo --output

report/SNTP_Assignment_02_Report.pdf --roll-number 23F3008 --student "Ahmad Fareed".

- Fixed seeds: data split 42, MF 42, ranking sample 42, diffusion base 42, target randomization 4242, influence 42, intervention 42. Each stage records the exact seed schedule in its JSON.
- validation/test selection discipline: model, lambda and target choices use validation or training data only; test is read once for the final tables.

8.2 Limitations (all disclosed)

- Full-scale config not executed. configs/default.yaml is provided for a longer run but no full-scale result is claimed.
- Restricted influence pool. R5 cannot sweep all 9,883 users; the comparison holds inside the 25 highest out-degree candidates and the graph is near saturation.
- Simulated outcomes. Epinions has no adoption timestamps; every diffusion, intervention and satisfaction number is simulated. The rating proxy is model-predicted, not observed after an intervention.
- MF ranking trade-off. Coverage rose roughly 17× but test nDCG@10 fell ~8.5%; validation improved, so the model was kept and the trade-off is reported.
- Sampled evaluation. Ranking/diversity use a fixed 500-user sample; coverage and diversity are sample- and feature-dependent.
- Network null model. The randomization is one seeded degree-preserving rearrangement, not full mixing; observed and randomized graphs cannot share edge-level thresholds.
- Group results. All three strategies chose the same top item in these training-only groups, so their simulated spreads are identical.

8.3 Conclusion

The submission is reproducible and internally consistent: the recommender comparison favours the trust-aware social model on observed metrics; CRN-based simulated cascades are governed by probability and preference rather than by target popularity alone; the observed network arrangement beats a degree-preserving randomization only in the sparse-propagation regime; greedy improves on degree only marginally inside a restricted high-degree pool; and CELF reproduces greedy with about 27% fewer evaluations once common random outcomes make the objective deterministic. Weak and negative results are reported alongside the positive ones.

References

[1] P. Massa and P. Avesani, “Trust-aware recommender systems,” in Proc. ACM RecSys, 2007.

[2] D. Kempe, J. Kleinberg, and É. Tardos, “Maximizing the spread of influence through a social network,” in Proc. ACM KDD, 2003.

[3] J. Leskovec, A. Krause, C. Guestrin, C. Faloutsos, J. VanBriesen, and N. Glance, “Cost-effective outbreak detection in networks,” in Proc. ACM KDD, 2007.

[4] W. O. Kermack and A. G. McKendrick, “A contribution to the mathematical theory of epidemics,” Proc. R. Soc. A, vol. 115, 1927.

[5] Y. Koren, R. Bell, and C. Volinsky, “Matrix factorization techniques for recommender systems,” IEEE Computer, vol. 42, 2009.

[6] B. Sarwar, G. Karypis, J. Konstan, and J. Riedl, “Item-based collaborative filtering recommendation algorithms,” in Proc. WWW, 2001.

[7] J. L. Herlocker, J. A. Konstan, L. G. Terveen, and J. Riedl, “Evaluating collaborative filtering recommender systems,” ACM TOIS, vol. 22, 2004.

[8] M. Jamali and M. Ester, “A matrix factorization technique with trust propagation for recommendation in social networks,” in Proc. ACM RecSys, 2010.